MASTER BLASTER QUESTION BANK

FOR JEE I NEET I CUET I BOARD & COMPETITIVE EXAMS

LAWS OF MOTION

PART-I SOLVED QUESTION ANSWER SERIES

SOLVED QUESTION ANSWER SERIES

Ex. 1 A boy standing on a weighing machine observes his weight as 200N. When he suddenly jumps upwards, his friend notices that the reading increased to 400N. The acceleration by which the boy jumped will be-

Sol. Force causing the acceleration

$= 400 - 200 = 200N$

mass of the boy = 200/9.8

hence acceleration = F/m

$$= \frac{200}{200} \times 9.8 = 9.8 \text{ m/s}^2$$

Ex. 2 A force of $(6\hat{i} + 8\hat{j})$ N acted on a body of mass 10kg. The displacement after 10sec, if it starts from rest, will be -

Sol. Acceleration $= \frac{\vec{F}}{m} = \frac{6\hat{i}+8\hat{j}}{10}$ in the direction of force and displacement

$$\vec{S} = \vec{u}t + \frac{1}{2}\vec{a}t^2$$

$$= 0 + \frac{1}{2}\left(\frac{6\hat{i}+8\hat{j}}{10}\right) 100 = 30\hat{i} + 40\hat{j}$$

So the displacement is 50m along $\tan^{-1}\frac{4}{3}$ with x-axis

Ex. 3 A fireman has to carry an injured person of mass 40kg from the top of a building with the help of the rope which can withstand a load of 100kg. The acceleration of the fireman if his mass is 80kg, will be-

Sol. Total mass = 80 + 40 = 120 kg

The rope cannot with stand this load so the fire man should slide down the rope with some acceleration

$\therefore$ The maximum tension = 100 × 9.8N

m (g – a) = tension,

$120 (9.8 - a) = 100 \times 9.8 \Rightarrow a = 1.63 \text{m/s}^2$

Ex. 4 A body of 0.02 kg falls from a height of 5 m into a pile of sand. The body penetrates the sand a distance of 5 cm before stoping. What force has the sand exerted on the body.

Sol. Suppose the velocity of the body at the instant when it reaches the pile of sand be v. Then

$v^2 = 0 + 2(9.8) \times (5 \text{ metre}) = 98$

$(\because v^2 = u^2 + 2as)$

$$a = -\frac{98}{2\times(0.05)} = -980 \text{ m/sec}^2$$

Now, retarding force

F = mass × acceleration

$= 0.02 \text{ kg} \times (-980 \text{ m/sec}^2) = -19.6 \text{ N}$

Ex. 5 The total mass of an elevator with a 80kg man in it is 1000kg. This elevator moving upward with a speed of 8 m/sec, is brought to rest over a distance of 16m. The tension T in the cables supporting the elevator and the force exerted on the man by the elevator floor will respectively be-

Sol. (a) The elevator having an initial upward speed of 8 m/sec is brought to rest within a distance of 16m. Hence,

$0 = (8)^2 + 2a(16)$ $(\because v^2 = u^2 + 2as)$

,

$$a = -\frac{8\times 8}{2\times 16} = -2 \text{ m/sec}^2.$$

Resultant upward force on elevator = T – mg.

According to Newton's law.

$T - mg = ma$

or $T = mg + ma = m(g + a)$

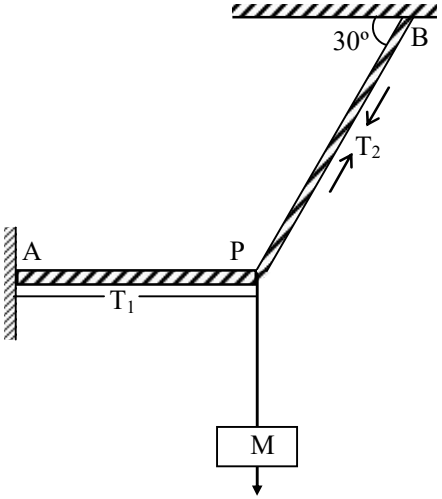
$= 1000 (9.8 - 2) = 7800N.$

(b) Let P be the upward force exerted on the man by the elevator floor, If m' be the mass of the man, then, Weight of the man acting downward = m' g , Upward force on the man

$= P - m'g$

According to Newton's law. P – m' g = m' a or P = m' (a + g) = (–2 + 9.8) = 624N.

Ex. 6 A mass M is hung with a light inextensible string as shown in figure. Find the tension horizontal string will be –



Sol. As there is a load at P so tension in AP and PB will be different.

Let these be T_1 and T_2 respectively. For vertical equilibrium of P

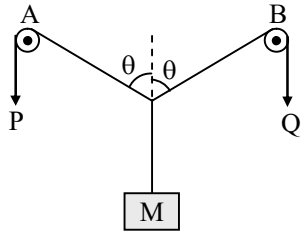
$T_2 \cos 60^\circ = Mg \Rightarrow T_2 = 2Mg$ ..... (1)

and for horizontal equilibrium of P

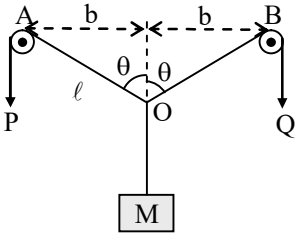
$T_1 = T_2 \sin 60^\circ = T_2 (\sqrt{3}/2)$ ..... (2)

Substituting the value of T_2 from equation(1) in (2) $T_1 = (2Mg) \times \sqrt{3}/2 = \sqrt{3}Mg$

Ex. 7 In the arrangement shown in fig the ends P and Q of an unstretchable string move downwards with uniform speed U. Pulleys A and B are fixed. Mass M moves upwards with a speed



Sol. As P and Q move down, the length l decreases



at the rate of Um/s.

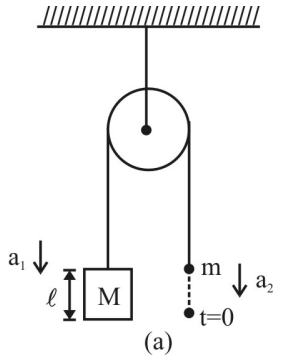
From figure, $l^2 = b^2 + y^2$

Differentiating with respect to time

$$2l \frac{d\ell}{dt} = 2y \frac{d\ell}{dt} \quad (\because \text{ b is constant})$$

$$\therefore = \frac{dy}{dt} = \frac{\ell}{y} \cdot \frac{d\ell}{dt} = \frac{1}{\cos\theta} \cdot \frac{d\ell}{dt} = \frac{U}{\cos\theta}$$

Ex. 8 Fig shows an arrangement of a rod of length l and mass M and a bead of mass m attached to a weightless string passing over a frictionless pulley. At t = 0, the bead is in level with the upper end of the rod. The bead slides down the string with considerable friction, and is opposite to the other end of the rod after t second. Assuming friction between the bead and the string to be constant all through, the frictional force is



Sol. Let a_1 and a_2 be the acceleration of M and m respectively .

Then $Mg - F = Ma_1$ .....(1)

and $mg - F = ma_2$ .....(2)

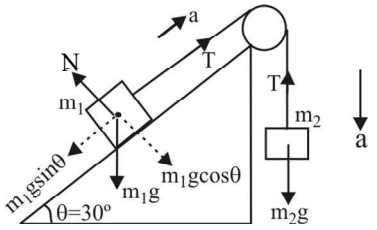
Now $l + \frac{1}{2}a_2t^2 = \frac{1}{2}a_1t^2$ or $a_1 = \frac{2\ell}{t^2} + a_2$

Solving equation. (1), (2) &(3) we get

$F = 2Mml/(M - m)t^2$

Ex.9 A block of mass m_1 = 4kg on a smooth inclined plane of 30º is connected by a cord over a small, frictionless pulley to a second block of mass m_2 = 5kg hanging vertically. The acceleration with which the block moves and the tension in the cord will respectively be- (Take g = 10m/sec^2)

Sol. The different forces acting on the masses are shown in fig



We have $T - m_1 g \sin\theta = m_1 a$ ...(1)

and $m_2 g - T = m_2 a$ ...(2)

Solving equations (1) and (2),

we get $a = \dfrac{(m_2 - m_1 \sin\theta)}{(m_1 + m_2)} g$ ...(3)

and $T = m_2 g\left[1 - \dfrac{(m_2 - m_1 \sin\theta)}{(m_1 + m_2)}\right]$

or $T = \dfrac{m_1 m_2 (1 + \sin\theta) g}{(m_1 + m_2)}$ ....(4)

Here m_1 = 4kg, m_2 = 5kg, θ = 30º

and g = 10m/s^2

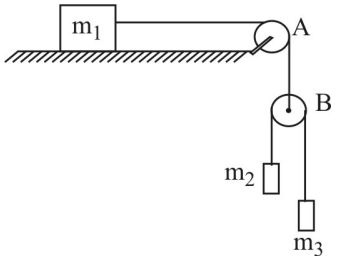
Substituting these values in equation (3), we get

$$a = \frac{[5 - (4 \times \frac{1}{2})] \times 10}{(4+5)} = \frac{30}{9} = 3.3 \text{m/sec}^2$$

Further substituting the values in eq. (4), we get

$$T = \frac{4 \times 5(1 + \frac{1}{2}) \times 10}{(4+5)} = \frac{300}{9} = 33.33\text{N}.$$

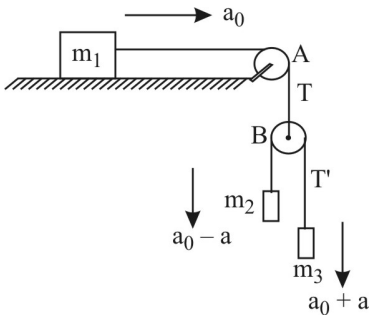
Ex. 10 Three blocks of mass m_1, m_2 and m_3 are connected as shown in the figure. All the surfaces are frictionless and the string and the pulleys are light. Find the acceleration of m_1.



Sol. Suppose the acceleration of m_1 is a_0 towards right. That will also be the downward acceleration of the pulley B because the string connecting m_1 and B is constant in length. Also the string connecting m_2 and m_3 has a constant length. This implies that the decrease in the separation between m_2 and B equals the increase in the separation between m_2 and B. So, the upward acceleration of m_2 with respect to B equals the downward acceleration of m_3 with respect to B. Let this acceleration be *a*.

The acceleration of m_2 with respect to the ground = a_0 – a (downward) and the acceleration of m_3 with respect to the ground = a_0 + a (downward).

These accelerations will be used in Newton's laws. Let the tension be T in the upper string and T' in the lower string. Consider the motion of the pulley B.



The forces on this light pulley are

(a) T upwards by the upper string, and

(b) 2T ′ downward by the lower strings.

As the mass of the pulley is negligible,

$2T' - T = 0$

giving $T' = T/2$ ....(i)

Motion of m_1 :

The acceleration is a_0 in the horizontal direction. The forces on m_1 are

(a) T by the string (horizontal)

(b) m_1g by the earth (vertically downwards) and

(c) N by the table (vertically upwards)

In the horizontal direction, the equation is

$$T = m_1a_0 \qquad \text{....(ii)}$$

Motion of m_2 : acceleration is $a_0 - a$ in the downward direction. The forces on m_2 are

(a) m_2g downward by the earth and

(b) $T' = T/2$ upward by the string

Thus , $m_2g - T/2 = m_2(a_0 - a)$ ....(iii)

Motion of m_3 : The acceleration is $(a_0 + a)$ downward. The forces on m_3 are

(a) m_3g downward by the earth and

(b) $T' = T/2$ upward by the string. Thus

$m_3g - T/2 = m_3 (a_0 + a)$ ....(iv)

We want to calculate a_0, so we shall eliminate T and a from (ii), (iii) and (iv)

Putting T from (ii) in (iii) and (iv)

$$a_0 - a = \frac{m_2g - m_1a_0/2}{m_2} = g - \frac{m_1a_0}{2m_2} \text{ and}$$

$$a_0 + a = \frac{m_3g - m_1a_0/2}{m_3} = g - \frac{m_1a_0}{2m_3}$$

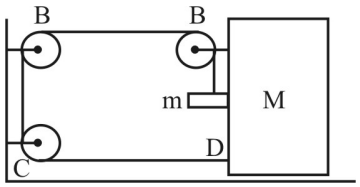
$$\text{Adding } 2a_0 = 2g - \frac{m_1a_0}{2}\left(\frac{1}{m_2} + \frac{1}{m_3}\right)$$

$$\text{or} \qquad a_0 = g - \frac{m_1a_0}{4}\left(\frac{1}{m_2} + \frac{1}{m_3}\right)$$

$$\text{or} \qquad a_0\left[1 + \frac{m_1}{4}\left(\frac{1}{m_2} + \frac{1}{m_3}\right)\right] = g$$

$$\text{or} \qquad a_0 = \frac{g}{1 + \frac{m_1}{4}\left(\frac{1}{m_2} + \frac{1}{m_3}\right)}$$

Ex. 11 In figure find the acceleration of m assuming that there is no friction between m and M, and all other surfaces are smooth and pulleys light. .



Sol. Let X' be the leftward displacement of A and x and y be the leftward and downward displacements of m. Then by constraint

$x = X \Rightarrow x = X \Rightarrow a_x = A_x$

and $l_1 - x + l_2 + l_3 - x + l_4 + y$

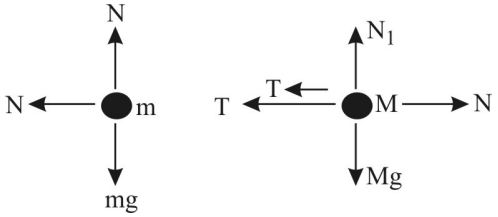
$= l_1 + l_2 + l_3 + l_4$

where l_1, l_2, l_3, l_4 are the instantaneous lengths of the segments of the string

$\Rightarrow 2x = y \Rightarrow 2x = y \Rightarrow 2a_x = a_y$

$N = ma_x$ and $mg - T = ma_y$

and $2T - N = MA_x = Ma_x$

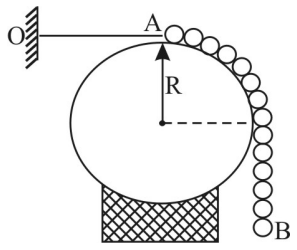


Eliminating T, A_x and N

$$a_x = \frac{2mg}{M + 5m} \text{ and } a_y = \frac{4mg}{M + 5m}$$

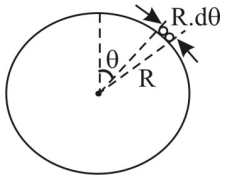
$$\therefore \qquad a = \sqrt{a_x^2 + a_y^2} = \frac{2\sqrt{5}mg}{M + 5m}$$

Ex. 12 A uniform flexible chain of length 1.25 m rests on a fixed smooth sphere of radius $R = 2/\pi$ m such that one end A of chain is at top of the sphere while the other end B is hanging freely. Chain is held stationary by a horizontal thread OA as shown in figure. Calculate acceleration of chain when the thread is cut ($g = 10\ ms^{-2}$)



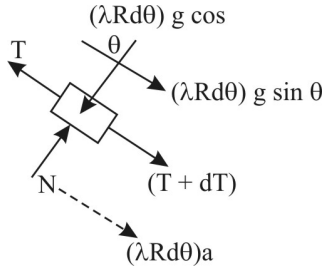
Sol. Top $\left(\frac{1}{2}\pi R\right)$ = 1 m length of chain is in contact with sphere and lower 0.25 m is hanging freely. When thread is cut, chain starts to accelerate. Every part of chain has same acceleration magnitude (top 1 m tangentially downwards and lower 0.25 m vertically downwards). Let the magnitude of acceleration be a and mass per unit length of chain be λ.

Considering free body diagram of an elemental length R . dθ (from top 1 m length) such that radius of sphere passing through this element is inclined at angle θ with vertical as shown in figure.



Mass of this elemental length = λ(Rdθ)

Considering tangential forces, (figure)



$(T + dT) + \lambda R g \sin\theta\, d\theta - T = \lambda R\, a\, d\theta$

or $dT = \lambda R (a - g \sin\theta)\, d\theta$

This equation is valid for top 1 m length of chain or from θ = 0 to θ = π/2.

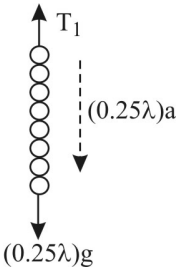
To solve above equation, integrate it,

$$\int_0^{T_1} dt = \lambda R \int_0^{\pi/2} (a - g \sin\theta)\, d\theta$$

(Note : θ = 0, means at top point A, tension is zero because thread is cut and tension is assumed to be T_1 at θ = π/2, means at junction of top 1 m length and lower 0.25 m length)

$$\therefore \quad T_1 = \lambda R \left(\frac{a\pi}{2} - g\right) \qquad .....(1)$$

Now considering free body diagram (figure) of lower 0.25 m length of chain,



$0.25\,\lambda g - T_1 = 0.25\,\lambda a$ ....(2)

From equations (1) and (2) a = 7.09 m/s^2